

Anatomic correction of transposition of the great vessels

We present a new approach for anatomic correction of transposition of the great arteries. The two coronary arteries, with a piece of the aortic wall attached, are transposed to the posterior artery. The two aortic openings are closed with a patch. The aorta and pulmonary artery are transected, contraposed, and then anastomosed. The interventricular septal defect is closed with a patch, through a right ventriculotomy approach, because the right ventricle is no longer part of the systemic circulation. Two patients, aged 3 months and 40 days and weighing 4,200 and 3,700 grams, respectively, were operated upon with deep hypothermia and total circulatory arrest. There was good recovery from the operation, with normal cardiocirculatory conditions. Renal failure developed in the first patient, and she died on the third postoperative day. During this time the cardiocirculatory conditions were good. The second patient made an uneventful recovery. Hemodynamic studies 20 days after the operation showed complete correction of the malformation. Five and one-half months after the operation, he weighs 7,500 grams, and his development is very good. We believe that this operation will be reproducible by most cardiovascular surgeons and will be an alternative to the Mustard procedure, especially for those patients with interventricular septal defect and pulmonary hypertension.

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The ideal operation for transposition of the great vessels must be one performed at the arterial level. To divide, contrapose, and then reanastomose the great arteries is not a surgical problem. The major technical difficulty in this approach has been the transfer of the coronary arteries. The earliest attempts made more than twenty years ago were uniformly unsuccessful.¹⁻⁵ Recently, anatomicopathological and experimental studies have been reported on this subject.⁶⁻⁸

This paper presents our initial experience with two infants subjected to anatomic correction of transposition of the great vessels. The case history of one of these patients has been related as a preliminary report.⁹

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Surgical procedure

The patients are operated upon with the use of profound hypothermia (16° C.) and total cardiocirculatory arrest according to the technique described by Mori,¹⁰ Barratt-Boyes,¹¹ and their colleagues.

The heart is exposed by means of a median sternotomy. The pericardium is opened wide by an inverted T-shaped incision. The ascending aorta and the pulmonary artery are dissected and freed distally to the level of the pericardial reflection and proximally to the level of the anterior and posterior valves. The initial portions of the coronary arteries are dissected free. With the heart beating normally, two sites are selected in the anterior wall of the posterior artery (in this case the pulmonary artery) where the coronary arteries will be located. These two places are identified with a stitch of 6-0 Prolene suture (Fig. 1, A). This is important, because when the heart is empty it is possible to open the artery in the wrong place. At the same time an area is selected in the aortic wall, adjacent to the origin of the coronary arteries, which will remain attached to the

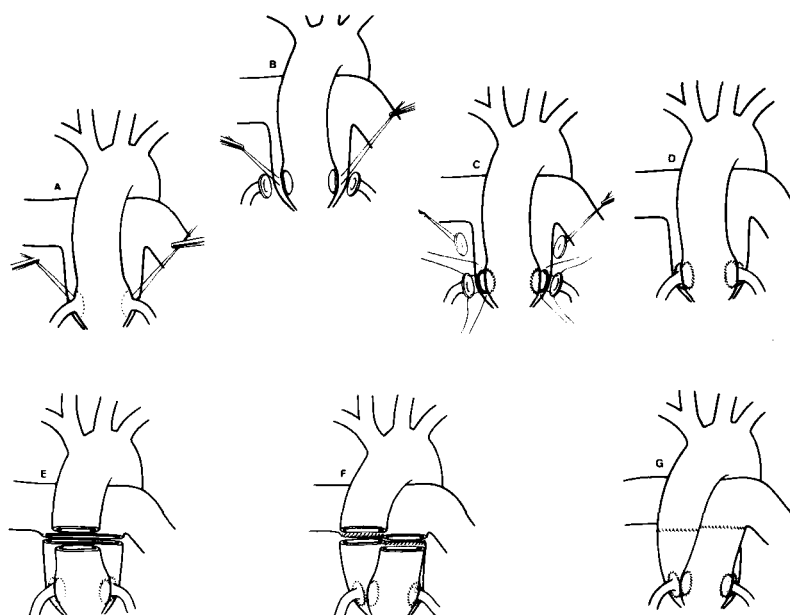


Fig. 1. Schematic drawing of the operative technique (see text).

coronary arteries. The transecting incisions in the aorta (anterior) and pulmonary artery (posterior) must be made at the same level and the sites selected at this time. The differences in diameter between the aorta and the pulmonary artery, instead of being a problem, are helpful when the two vessels are contraposed and reanastomosed. Purse-string sutures are placed in the right atrial appendage and at the apex of the left ventricle (in this case pulmonary ventricle), and cannulas are inserted. Another cannula is inserted in the ascending aorta through a small transverse aortotomy at the point selected for complete transection of the aorta. Perfusion is started and hypothermia to 16° C. is reached in 5 or 6 minutes. The blood is permitted to drain to the heart-lung machine, the aorta is cross-clamped just below the innominate artery, and the cannulas are removed from the aorta and right atrium. The coronary arteries are then excised with the selected piece of the aortic wall attached (Fig. 1, B). This maneuver simplifies the subsequent anastomosis. The openings in the aortic wall are closed with a running suture by means of a patch of homologous dura mater preserved in glycerol (Fig. 1, C). This is the same material used in the dura mater valve for cardiac valve replacement.¹² It is easier to suture the dura mater than the pericardium. With the two stitches previously placed as a guide, two similar pieces are resected from the posterior artery (in this case the pulmonary artery).

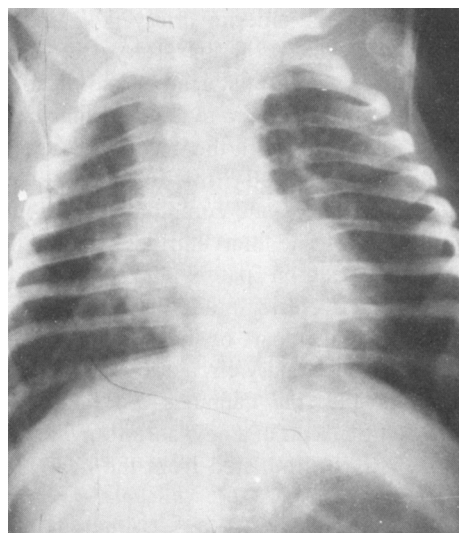


Fig. 2. Case 1. Preoperative roentgenogram.

Two anastomoses are then performed with continuous 6-0 Prolene suture, the coronary arteries being implanted in their new sites (Fig. 1, D). The next step is to transect the ascending aorta by using the aortotomy for cannulation. At the same level, the pulmonary artery is transected (Fig. 1, E). A continuous running suture is then placed in the distal end of the pulmonary artery starting at the right border. A similar suture is

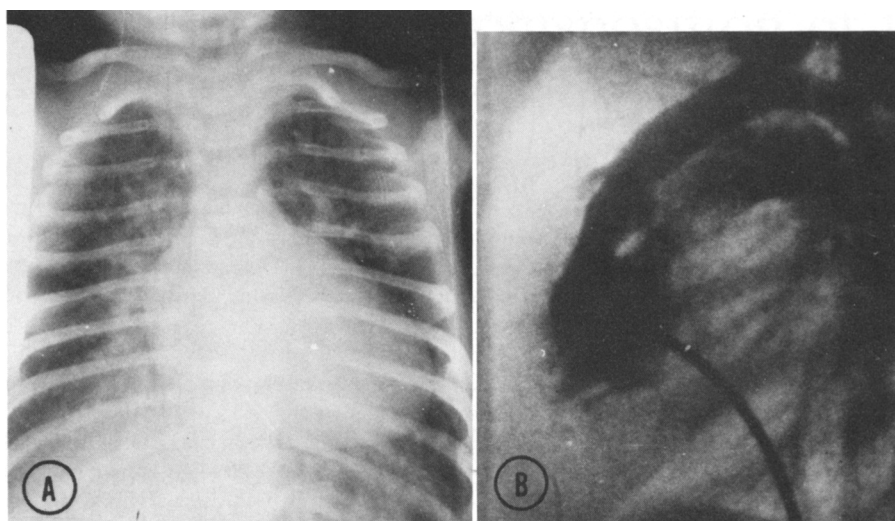


Fig. 3. Case 2. *A*, Preoperative roentgenogram. *B*, Preoperative cineangiogram.

placed at the proximal end of the pulmonary artery starting at the left border (Fig. 1, *F*). These two sutures are intended to equalize the diameters of the vessels which are going to be contraposed and reanastomosed. The distal end of the anterior artery is sutured to the proximal end of the posterior artery, to which the coronary arteries are now attached. The distal end of the posterior artery is sutured to the proximal end of the anterior artery, from which the coronary arteries were removed (Fig. 1, *G*). By this technique we have performed an anatomic and complete correction of the transposed vessels. The interventricular septal defect is closed with a Dacron patch, the approach being through a right ventriculotomy, because the right ventricle is no longer part of the systemic circulation but instead, the pulmonary circulation. After closure of the right ventricle, the cannulas are reinserted, the aortic line being placed in a new aortotomy distal to the suture line. Air is eliminated from the heart, and the patient is rewarmed.

The postoperative care for these infants is the same as that for any subjected to profound hypothermia.

Case reports

Two patients with transposition of the great vessels and interventricular septal defect were operated upon for anatomic correction.

CASE 1. C. P., an 11-day-old white female infant weighing 3,200 grams, was admitted with a history of cyanosis since birth. The clinical diagnosis was transposition of the great vessels. Cardiac catheterization at this time revealed a large interventricular septal defect and transposition of great

arteries. Balloon septostomy was not done because it was not possible to insert the catheter into the left atrium.

At the age of 3 months, her weight was 4,200 grams and she was becoming progressively more cyanotic. A second cardiac catheterization was then performed. Pressures were as follows: right ventricle 100/12 mm. Hg, aorta 100/60 mm. Hg, pulmonary artery 100/58 mm. Hg, and left ventricle 100/12 mm. Hg. The arterial oxygen saturation was 53 per cent. The electrocardiographic tracings showed right ventricular and right atrial hypertrophy. The chest roentgenogram demonstrated that the cardiac shadow was smaller than on the first film made, and there was an increased pulmonary vascularity, but without signs of congestion (Fig. 2).

The operative procedure described was performed on May 4, 1975. The only difference was in the closure of the two openings in the former aorta. The orifice on the left was closed with a pericardial patch and the one on the right with a continuous running suture. The complete correction, including closure of the interventricular septal defect through a right ventriculotomy, was done with the patient under total circulatory arrest. The operation took 75 minutes. After rewarming, the heart recovered completely with no conduction disturbances, and maintained good cardiocirculatory condition. Abnormal bleeding required extra stitches in the sutures of the patched coronary orifices.

The infant had a complete postoperative recovery. The intratracheal tube was removed after 6 hours and the infant was alert. The following day all conditions were good except for the urinary output. On the second day, anuria and elevations of the serum potassium and urea levels were noted. On the third day, before peritoneal dialysis was begun, she had a cardiac arrest and could not be resuscitated despite all attempts at that time. Necropsy showed all the sutures to be in place and the anatomic correction to be good.

CASE 2. P. C. O. B., a 40-day-old white male infant weighing 3,700 grams, was first examined on May 6, 1975. Cyanosis had first been noted 10 days after birth. On

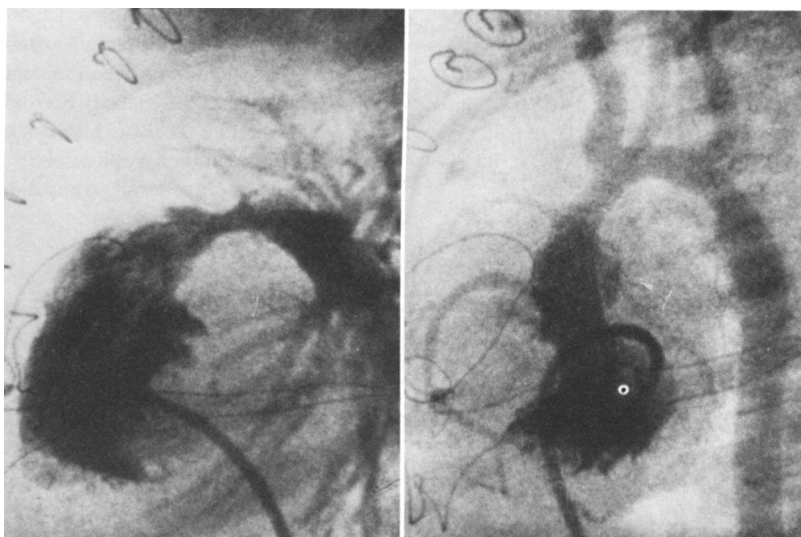


Fig. 4. Case 2. Postoperative cineangiogram.

admission he had a cough, tachypnea, and congestive heart failure with increasing cyanosis. He was treated with digitalis, diuretics, and antibiotics. On physical examination he was cyanotic and the liver was enlarged, the edge being 3 cm. below the right costal margin. There was a Grade 2 systolic murmur at the mesocardium. A roentgenogram revealed an enlarged, egg-shaped heart and a considerable increase in pulmonary vasculature (Fig. 3, A). Data obtained from cardiac catheterization showed a pressure of 90/13 mm. Hg in the right ventricle and 90/55 mm. Hg in the aorta. The pressure in the left ventricle was 90/12 mm. Hg and in the pulmonary artery, 75/35 mm. Hg. The arterial oxygen saturation was 88 per cent with the infant under general anesthesia. The cineangiogram confirmed the diagnosis of transposition of the great arteries with a large ventricular septal defect (Fig. 3, B).

The patient was operated upon on May 8, 1975, according to the technique described. The postoperative course was uneventful. Twenty days after the operation, right and left heart catheterization studies were performed. The cineangiographic aspect is shown in Fig. 4. We can see the complete correction and the coronary arteries at their new sites. The pressure in the pulmonary artery had dropped to 25/13 mm. Hg and that in the right ventricle to 60/10 mm. Hg. The mean pressures at the left and right atria were 10 and 6 mm. Hg, respectively. The infant was discharged 3 weeks after operation in good condition. He has been re-examined monthly and has had no problems. Ten months after the operation, without medications, he is thriving at normal standards and weighing 8,300 grams.

Comment

The idea of retransposing the great arteries in patients with transposition of the great vessels is not new. The attempts of Bailey and associates² and Kay and Cross³ and the proposal of Björk and Bouckaert¹³ leave the coronary arteries in the pulmonary artery. With the

technique of Mustard and associates¹ the left coronary artery is taken with the aorta but the right coronary artery remains in the pulmonary artery. Senning⁴ was the first who completely corrected the transposition, but his technique is very difficult to use. Idriss and associates⁵ in 1961 developed an interesting technique, but the ring of the aorta with the two coronary arteries is difficult to obtain and the anastomoses are all made very near the valve level. At this time all the patients in whom these techniques were tried have died in the operating room or shortly thereafter. Since that time we have not found new attempts in the literature. The papers of Anagnostopoulos,^{6, 7} Balderman,⁸ and their associates present interesting ideas but no clinical experience.

We based our technique on our experience with aorta-coronary bypasses,¹⁴ and it sounded technically feasible to us. There are two major arguments: First, the coronary arteries are excised with a piece of the aortic wall, so that there are no problems with suture and future stenosis; second, the great vessels are transected far from the valves, so that it is easier to create the anastomoses and to correct any leak that is observed. This technique also permits the adjustment of the sizes of the two vessels. The operation may be technically difficult, but we believe it is reproducible by most of the cardiovascular surgeons. We still believe at the present time that transposition of the great vessels with an intact ventricular septum must be corrected by the Mustard¹⁵ operation. We think in this situation that the left ventricle, having been exposed to a low pressure regimen, perhaps will not be able to

sustain the load of systemic pressure. The same is true in cases of pulmonary stenosis. The Rastelli¹⁶ operation should be the procedure of choice for this group of patients if the stenosis cannot be relieved properly. However, if the stenosis can be relieved, perhaps relief of stenosis will be adequate for anatomic correction. Our technique will be ideal for transposition of the great vessels with ventricular septal defect or persistent ductus arteriosus and pulmonary hypertension with low arteriolar resistance. We do not know at the present time whether the procedure is adequate for patients who have pulmonary hypertension and high arteriolar resistance.

Some successful similar operations have been performed by Kreutzer,¹⁷ Yacoub,¹⁸ Ross,¹⁹ and Nakamura.²⁰

Addendum

After this paper was submitted, we had a disappointing experience. Five infants operated upon died a few hours after the operation or in the operating room. In all these patients there were no technical problems. The first patient, 3 months old, had a severe respiratory infection for which he required assisted ventilation for a week before the operation. At that time he was resuscitated from a cardiac arrest, and the operation was a desperate effort to change the situation. The next 3 patients were between 2 and 3 months old. All of them had patent ductus arteriosus, and 2 of them also had a ventricular septal defect. Evidence of pulmonary hyperresistance was present. The heart size was relatively small. These infants did not have pulmonary vascular congestion or cardiac insufficiency, but they did have severe anoxia. After complete anatomic correction and closure of the patent ductus, the pressure in the pulmonary artery appeared higher than that in the aorta. Output was low. In one of these cases the pressure measured in the left ventricle was 65 mm. Hg and that in the right ventricle 80 mm. Hg. The end-diastolic pressure was high. Opening the ductus decreased both the end-diastolic pressure and the pressure difference between the two ventricles. Despite this maneuver, the patient died.

The fifth patient we considered to be a good subject for this operation. He had signs of low pulmonary resistance with an enlarged heart, congestion of the pulmonary vasculature, and heart failure. After the complete anatomic correction, the aortic pressure was 90/60 mm. Hg and the right ventricular systolic pressure was 40 mm. Hg. This patient had an excellent immediate recovery, with good cardiocirculatory condition and good urinary output. He was completely awake, and the blood gas and electrolyte values were almost normal. Six hours after the operation, he had a sudden cardiac arrest and could not be resuscitated.

Despite this very discouraging experience, we still believe that this technique will be useful, especially if we improve the selection of patients.

On last June 20, we performed another successful anatomic correction on a 9-month-old girl who had transposition of the great vessels and a large ventricular septal defect with signs of low pulmonary resistance. Because the pulmonary artery was to the right, a piece of Dacron tube was used to eliminate tension in the pulmonary flow pathway reconstruction. One month after the operation, the patient is doing remarkably well.

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Discussion

DR. ALBERT STARR

Portland, Ore.

Our own interest in anatomic correction of transposition was stimulated by this beautiful work by Dr. Jatene and began in October, 1975, after we heard of his successful operation. Our goal was to explore the application of anatomic correction to patients with intact septum. Our patients were 12 months, 5 months, and 11 months of age. The first two had simple transposition with ductus, no ventricular septal defect (VSD), and left ventricular pressures less than systemic. The last patient had easily correctible subpulmonic stenosis with systemic pressures in the left ventricle.

[Slide] The operation is done with a standard cardiopulmonary bypass, no surface cooling, and hypothermia without circulatory arrest. The aorta is cross-clamped during transfer of the coronary arteries, in the method described by Jatene, and during the restoration of systemic outflow continuity. It takes between 45 and 50 minutes of aortic cross-clamping to transfer both coronary arteries and join the distal aortic segment to the proximal pulmonary artery segment.

The anatomy following repair is shown here [slide]. Note that tailoring of the proximal pulmonary artery segment places the left coronary artery to the left of the reconstructed pulmonary outflow tract and out of the danger of kinking or compression.

Our first patient had an excellent postoperative course. There was no evidence of circulatory problems despite an intact ventricular septum preoperatively.

On the tenth day the patient was noted to have an infected cutdown in the neck which grew *Serratia*, and the patient

subsequently died of a septicemia. However, this patient had excellent circulation for the first 10 days after the operation, without evidence of hypotension, despite the fact that the left ventricular pressure prior to operation was significantly less, about half the systemic pressures.

Our second and third patients did not survive, and we felt this was due to kinking and compression of the left coronary system by tension of the proximal aorta-pulmonary artery anastomosis on the left coronary system. The experience with our first case suggests to us that anatomic correction may be applicable to patients with intact septum.

In addition, we have learned from this experience that tension must be avoided under the reconstructed pulmonary flow pathway to avoid injury to the left coronary artery. This can be accomplished by cutting the proximal aortic stump long and, as in our first case and as in Dr. Jatene's cases, by tailoring the proximal pulmonary artery so that the left coronary anastomosis is placed far to the left of the aortic stump, where it cannot be compressed by the overlying reconstructive right ventricular outflow tract.

I think this is a very important area for further explanation. Dr. Jatene's surviving patient and perhaps others, as well, serve as an impetus to us to explore further the anatomic correction of transposition.

DR. HUGHE WILSON III

Dallas, Texas

I rise to congratulate Dr. Jatene and to raise a question which I think needs to continue to trouble us. That is, at what narrow limits of elevated pulmonary vascular resistance can this arterial retransposition procedure be used which might yield survival figures that can compete with our results with the intra-atrial baffling procedures? To illustrate, without apology, I use slides that we used to illustrate this point 16 years ago.

[Slide] Many have attempted arterial retransposition, starting with Björk who reported his efforts without bypass in 1954, Kay and Cross in 1955, Bailey with hypothermia in 1954, Mustard with the left coronary artery in 1954, Crafoord and Senning with both coronary arteries in 1955, and others who did not report their failures. All of these attempts ended disastrously. Most of the patients died in low output states with pulmonary edema or renal failure or both.

In 1960, at Dr. Kirklin's request, we discussed transposition of the great vessels at the meeting of the American College of Surgeons. We pointed out that after the first month or two of life when the fetal vascular resistance begins to fall, the left ventricle becomes weaker than the right except in patients with fixed pulmonary hypertension. If the fetal pulmonary vascular resistance has fallen, and we are pitting a relatively hypoplastic left ventricle against the systemic resistance, we can expect left ventricular failure, pulmonary edema, and death as described today. On the other hand, if Class IV pulmonary vascular changes have developed, we would not expect survival with the intracardiac repair of even much less complex lesions.

We need now to find out at which point the left ventricular hypertrophy is sufficient to allow these patients to survive or else, after 22 years, we must accept the physiological limitations.

DR. YASUNARU KAWASHIMA

Osaka, Japan

[Slide] I would like to congratulate Dr. Jatene on his excellent presentation and his superb technique. I also have done this procedure recently in an infant, with success. In this case, I transected the pulmonary artery obliquely from left proximal to right distal.

With this technique, the length of the new great arteries increased somewhat, and the anastomosis was easier to perform than when the pulmonary artery is transected at right angles. The patient survived and is well now.

In another patient, I transected the pulmonary artery in the opposite fashion, from right proximal to left distal. With this approach, the transecting line can be more oblique and, therefore, the new great arteries are lengthened further. The new great arteries were anastomosed in the opposite way to that reported by Dr. Jatene—that is, the new aorta was anastomosed on the left and the new pulmonary artery on the right.

This patient did not survive the operation, but the method of anastomosis was probably not the reason for his death. I do not know which type of anastomosis is more desirable, but at least technically the latter way was easier.

DR. RICHARD R. LOWER

Richmond, Va.

Most of the pertinent points have been covered by the other discussers. I just want to add one word of encouragement with regard to the concept that Dr. Jatene proposed, and that is the feasibility of doing the procedure in an infant who does *not* have a large ventricular septal defect and pulmonary hypertension.

We operated upon a 7 pound infant a few months ago who had an intact septum and no pulmonary hypertension. The procedure seemed relatively simple, and there was absolutely no difficulty with left ventricular function.

The one technical feature that I learned is the advisability of transecting the aorta, as Dr. Starr mentioned, well out from the heart, so that it loops around to reach the distal pulmonary artery without much kinking or tension.

This particular infant did make an uneventful recovery from the cardiovascular standpoint, and there was not this problem of the underdeveloped left ventricle. The infant continues to do well.

The ultimate value and potential of the procedure, at least in part, will rest on the growth potential of these coronary button anastomoses. Now that the technique has been shown

to be feasible clinically, we are taking the problem to the experimental laboratory in hopes of clarifying this particular point.

DR. KEITH W. ASHCRAFT

Kansas City, Mo.

Dr. Jatene's accomplishment is truly a milestone in the management of congenital heart disease, one which Eoin Aberdeen likened to the conquest of Mt. Everest.

Dr. Tom Holder and I have done five arterial switch operations for transposition of the great vessels in infants with complex transposition malformations. All were desperately ill, and we assessed their chances as *very* poor by more standard techniques. We used a modification of the Baffes technique and learned very early, as Dr. Starr has suggested, that it is necessary to cut the aorta well away from the heart so that as the pulmonary artery is brought around for repositioning it will not compress the coronary arteries against the myocardium.

Dr. Jatene has made another important alteration in the Baffes technique—transecting both great vessels well away from the heart and moving the coronary arteries with a button of aorta into the new left ventricular outflow tract. In our patients some bleeding has occurred at the suture line where the coronary arteries have been included in the newly positioned aorta. Since it is at the base of the heart, access for hemostasis is virtually impossible.

One of our patients lived 13 hours, but the other 4 did not survive the operating room.

I think hypothermic arrest is important, as Dr. Jatene suggests, because of its simplicity. Again, transecting both great vessels well away from the heart is very important.

DR. J. ATENE (*Closing*)

I would like to thank the discussers for their comments and to congratulate Dr. Kawashima and Dr. Lower for their successful cases. I agree with Dr. Starr and Dr. Ashcraft. There are many critical details in this technique and tension must be avoided, especially under the reconstructed pulmonary flow pathway. Dr. Yacoub and Dr. Donald Ross have inserted a piece of homograft or Dacron tube to obviate all this inconvenience. I think this is a very interesting point, especially in those cases in which the pulmonary artery is at the right. About Dr. Wilson's question, I really don't know at which point left ventricular hypertrophy is sufficient to allow these patients to survive. Regarding pulmonary vascular resistance, the patient is not candidate for surgical correction if it is significantly elevated. All 4 patients operated upon in this situation died. Those patients with high pressure in the left ventricle and with normal or easily correctable outflow tract and low pulmonary vascular resistance should be the best candidates for this procedure.